

Kelsey Lawson

United States Citizen
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EDUCATION

University Of California - Irvine, Irvine, California

September 2017 - June 2021

Bachelor of Science in Materials Science and Engineering, GPA 3.192/4.00

Minor in Biomedical Engineering & Specialization in Mechanical Design

Dean's List Winter Quarter 2020 & 2021

CERTIFICATES

Certified Solidworks Associate (CSWA)

Certificate ID: C-2VJYA6B7XS

SKILLS

- **Computer:** SolidWorks, MATLAB, Arduino Uno, Microsoft Word, PowerPoint, Excel, and Google Docs, Sheets, Slides.
- **Laboratory:** scanning electron microscopy (SEM), transmission electron microscopy (TEM), optical imaging analysis, x-ray diffraction (XRD) graph analysis, tensile, compression, and bending tests, stress analysis, Raman spectroscopy analysis, and electrospinning.

RELEVANT COURSEWORK

- | | | | |
|------------------------|---|----------------------------------|--------------------------|
| • Materials Processing | • Cell, and Molecular Engineering | • Sensory-Motor Systems | • Nanomaterials |
| • Mechanical Systems | • Polymers & Synthetic Materials | • Statics/Lightweight Structures | • Theory of Machines |
| • Design Failure | • Electronic Optical Properties | • Organ Transport Systems | • Composite Structures |
| • Dynamics | • Micro-Electromechanical System (MEMS) | • Thermodynamics of Materials | • Diffusion in Materials |

ACADEMIC PROJECTS

SiPhox, Inc - Senior Design Project, Irvine, California

September 2020 - June 2021

Design Engineer & Programmer

- Collaborated with SiPhox Inc., a small startup company backed by Y Combinator worth \$5.8 million to design a multipurpose prototype of a mechanical system to keep fluid samples agitated at a constant temperature for efficient mixing.
- Developed 3D models and engineering drawings to showcase the expected design and monitored rapid prototyping processes and conducted first article inspection (FAI) for visual and quality inspection.
- Implemented Arduino program to complete the prototype's motor design before the deadline and under the designated budget of \$2500.
- Applied lean six sigma tools to maximize analysis of system, design, process, service, and safety FMEA.

Biomedical Engineering Design Systems, Irvine, California

January 2020 - December 2020

Detextroke - Stroke Detection Device

Research Engineer

- Explored how electroencephalograms (EEG) analyze brain activity to create a theoretical system for a stroke patient with little to no verbal and motor functions to reach emergency assistance through brain signal recognition.
- Identified relevant questions to determine the best method to use EEG in a theoretical system.
- Targeted specific demographic parameters on stroke patients with little to no verbal and motor functions, which delivered emergency assistance through EEG signal recognition.

3D Printed Elbow External Fixator Device

Design Engineer & Material Researcher

- Delivered effective presentations outlining the prototype material research and assembly.
- Designed engineering drawings and assembly using SolidWorks.
- Completed prototype consistently sticking to a strict 10-week schedule and budget of \$215 from start to finish while adapting to schedule changes from the pandemic.

Gaming Controller For A Virtual Robot Vehicle, Irvine, California

April 2020 - June 2020

Design Engineer & Programmer

- Supported engineering design development through analysis and simulation of prototypes and 3D computer models.
- Built a functional prototype of a fully steerable virtual Agar.io vehicle using Arduino to meet design specifications.
- Prepared, checked and coordinated documentation to support component design and application.

Biomaterials & Biomimetics Research Lab, Irvine, California

January 2020 - March 2020

Research Assistant

- Collected, prepared, and tested a wide range of biological samples.
- Stocked, organized, and prepared supplies inventory.
- Documented procedures and results with a high degree of accuracy and precision.

WORK EXPERIENCE

The Home Depot, Roseville, California

June 2017 - September 2018

Cashier

- Restocked, arranged, and organized merchandise in front lanes to drive product sales.
- Mentored new team members on POS system operation, customer service strategies, and sales goals.
- Helped customers complete purchases, locate items and join reward programs to promote loyalty, satisfaction, and sales numbers.

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MATERIALS SCI ENGR (SCHOOL OF ENGINEERING)

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University Requirements

09/01/17 Entry Level Writing - Satisfied
09/01/17 American History - Satisfied
09/01/17 American Institutions - Satisfied

Bachelor's Degrees

BS MATERIALS SCIENCE ENGINEERING
JUNE 11, 2021
06/11/21 MINOR IN BIOMEDICAL ENGINEERING

Memoranda

02/15/18 MAJOR CHANGED FROM

* ENGR / UNDECIDED

AP ENGLISH LIT (Score 3, Units 8.0) 05/17

AP CALCULUS AB (Score 3, Units 4.0) 05/17

AP ENGLISH LANG (Score 3, Units 0.0) 05/16

Units Transferred 12.0

2017 Fall Quarter

INTRO ENGR COMP	ENGRMAE	10	4.0	C	8.0
GENOCIDE SINCE WWII	HISTORY	11	4.0	A	16.0
GENERAL CHEMISTRY	CHEM	1A	4.0	B-	10.8
CALCULUS	MATH	2A	4.0	A-	14.8
Term Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 49.6	GPA: 3.100	BAL: 17.6
Cumulative Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 49.6	GPA: 3.100	BAL: 17.6

2018 Winter Quarter

GENERAL CHEMISTRY	CHEM	1B	4.0	B-	10.8
CALCULUS	MATH	2B	4.0	B	12.0
CLASSICAL PHYSICS	PHYSICS	7C	4.0	A-	14.8
CLASS PHYSICS LAB	PHYSICS	7LC	1.0	A	4.0
ENVIRON HAZARDS	PP&D	142	4.0	A	16.0
Term Totals	ATTM: 17.0	PSSD: 17.0	GPTS: 57.6	GPA: 3.388	BAL: 23.6
Cumulative Totals	ATTM: 33.0	PSSD: 33.0	GPTS: 107.2	GPA: 3.248	BAL: 41.2

2018 Spring Quarter

GENERAL CHEMISTRY	CHEM	1C	4.0	C+	9.2
GENRL CHEMISTRY LAB	CHEM	1LC	2.0	B+	6.6
MULTIVAR CALCULUS	MATH	2D	4.0	C	8.0
CLASSICAL PHYSICS	PHYSICS	7D	4.0	C	8.0
CLASS PHYSICS LAB	PHYSICS	7LD	1.0	C	2.0
Term Totals	ATTM: 15.0	PSSD: 15.0	GPTS: 33.8	GPA: 2.253	BAL: 3.8
Cumulative Totals	ATTM: 48.0	PSSD: 48.0	GPTS: 141.0	GPA: 2.938	BAL: 45.0

2018 Fall Quarter

PRIN MATERIALS SCI	ENGR	54	4.0	B+	13.2
ENVIRON CHALLENGES	ENGRCEE	60	4.0	A	16.0
INTRO LINEAR ALGBRA	MATH	3A	4.0	B	12.0
CLASSICAL PHYSICS	PHYSICS	7E	4.0	C-	6.8
Term Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 48.0	GPA: 3.000	BAL: 16.0
Cumulative Totals	ATTM: 64.0	PSSD: 64.0	GPTS: 189.0	GPA: 2.953	BAL: 61.0

2019 Winter Quarter

STATICS	ENGR	30	4.0	C+	9.2
THERMO OF MATERIALS	CBEMS	65A	4.0	A-	14.8
CRIT READING & RHET	WRITING	39B	4.0	A+	16.0
ELEM DIFF EQUATIONS	MATH	3D	4.0	B-	10.8
Term Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 50.8	GPA: 3.175	BAL: 18.8
Cumulative Totals	ATTM: 80.0	PSSD: 80.0	GPTS: 239.8	GPA: 2.998	BAL: 79.8

2019 Spring Quarter

DIFFUS IN MATERIALS	CBEMS	65B	4.0	C+	9.2
NANOMATERIALS	CBEMS	141	4.0	B	12.0
NETWORK ANALYSIS I	EECS	70A	4.0	C+	9.2
MULTIVAR CALCULUS	MATH	2E	4.0	C	8.0
Term Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 38.4	GPA: 2.400	BAL: 6.4
Cumulative Totals	ATTM: 96.0	PSSD: 96.0	GPTS: 278.2	GPA: 2.898	BAL: 86.2

2019 First Summer Session

CELL MOLECULAR ENGR	BME	50A	4.0	B+	13.2
Term Totals	ATTM: 4.0	PSSD: 4.0	GPTS: 13.2	GPA: 3.300	BAL: 5.2
Cumulative Totals	ATTM: 100.0	PSSD: 100.0	GPTS: 291.4	GPA: 2.914	BAL: 91.4

2019 Special / 10-Week Summer Session

ARGUMENT & RESEARCH	WRITING	39C	4.0	A-	14.8
Term Totals	ATTM: 4.0	PSSD: 4.0	GPTS: 14.8	GPA: 3.700	BAL: 6.8
Cumulative Totals	ATTM: 104.0	PSSD: 104.0	GPTS: 306.2	GPA: 2.944	BAL: 98.2

2019 Second Summer Session

CELL MOLECULAR ENGR	BME	50B	4.0	A-	14.8
Term Totals	ATTM: 4.0	PSSD: 4.0	GPTS: 14.8	GPA: 3.700	BAL: 6.8
Cumulative Totals	ATTM: 108.0	PSSD: 108.0	GPTS: 321.0	GPA: 2.972	BAL: 105.0

2019 Fall Quarter

MUSIC IN MULTIMEDIA	MUSIC	46	4.0	A+	16.0
MECHANICS OF STRCTR	ENGR	150	4.0	B-	10.8

MECH OF STRUCT LAB	ENGRMAE	150L	1.0	A-	3.7
MAT KIN & PHASE TRN	ENGRMSE	165	3.0	B-	8.1
INTRO RACEDNICITY	CHC/LAT	64	4.0	A-	14.8
Term Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 53.4	GPA: 3.338	BAL: 21.4
Cumulative Totals	ATTM: 124.0	PSSD: 124.0	GPTS: 374.4	GPA: 3.019	BAL: 126.4
2020 Winter Quarter					
FUND OF MUSIC	MUSIC	25	4.0	A-	14.8
WAGNER	MUSIC	41	4.0	A	16.0
EAD:CAD	BME	60C	4.0	B+	13.2
MATLS PROCESSING	ENGRMSE	195	4.0	A-	14.8
DEANS HONOR LIST					
Term Totals	ATTM: 16.0	PSSD: 16.0	GPTS: 58.8	GPA: 3.675	BAL: 26.8
Cumulative Totals	ATTM: 140.0	PSSD: 140.0	GPTS: 433.2	GPA: 3.094	BAL: 153.2
2020 Spring Quarter					
MECH SYSTEMS LAB	ENGRMAE	106	4.0	A	16.0
POLYMER SCI & ENGR	ENGRMSE	154	4.0	B	12.0
MECH BEH/DES PRIN	ENGRMSE	155	4.0	B-	10.8
MECH BEHAVIOR LAB	ENGRMSE	155L	1.0	A-	3.7
XRD SEM &MICROANLYS	ENGRMSE	164	3.0	A-	11.1
XRD & MICROANLS LAB	ENGRMSE	164L	2.0	A-	7.4
Term Totals	ATTM: 18.0	PSSD: 18.0	GPTS: 61.0	GPA: 3.389	BAL: 25.0
Cumulative Totals	ATTM: 158.0	PSSD: 158.0	GPTS: 494.2	GPA: 3.128	BAL: 178.2
2020 First Summer Session					
DISNEYLAND	ART HIS	55	5.0	A	20.0
Term Totals	ATTM: 5.0	PSSD: 5.0	GPTS: 20.0	GPA: 4.000	BAL: 10.0
Cumulative Totals	ATTM: 163.0	PSSD: 163.0	GPTS: 514.2	GPA: 3.155	BAL: 188.2
2020 Special / 10-Week Summer Session					
COMPUTER AIDED DESN	ENGRMAE	52	4.0	B+	13.2
Term Totals	ATTM: 4.0	PSSD: 4.0	GPTS: 13.2	GPA: 3.300	BAL: 5.2
Cumulative Totals	ATTM: 167.0	PSSD: 167.0	GPTS: 527.4	GPA: 3.158	BAL: 193.4
2020 Second Summer Session					
COMM IN PROF WORLD	ENGR	190W	4.0	A	16.0
DYNAMICS	ENGRMAE	80	4.0	B-	10.8
Term Totals	ATTM: 8.0	PSSD: 8.0	GPTS: 26.8	GPA: 3.350	BAL: 10.8
Cumulative Totals	ATTM: 175.0	PSSD: 175.0	GPTS: 554.2	GPA: 3.167	BAL: 204.2
2020 Fall Quarter					
SENSORY MOTOR SYS	BME	120	4.0	B+	13.2
MEMS	EECS	179	4.0	A-	14.8
SR DESIGN PROJ I	MSE	189A	3.0	A-	11.1
Term Totals	ATTM: 11.0	PSSD: 11.0	GPTS: 39.1	GPA: 3.555	BAL: 17.1
Cumulative Totals	ATTM: 186.0	PSSD: 186.0	GPTS: 593.3	GPA: 3.190	BAL: 221.3
2021 Winter Quarter					
INTRO BIOMED ENGR	BME	1	3.0	A-	11.1
LIGHTWEIGHT STRUCTR	ENGRMAE	157	4.0	A-	14.8

LAB SYNTH MATLS	MSE	60	4.0	A	16.0
DESGN FAILURE INVST	MSE	175	4.0	B+	13.2
SR DESIGN PROJ II	MSE	189B	3.0	A-	11.1

DEANS HONOR LIST

Term Totals	ATTM: 18.0	PSSD: 18.0	GPTS: 66.2	GPA: 3.678	BAL: 30.2
Cumulative Totals	ATTM: 204.0	PSSD: 204.0	GPTS: 659.5	GPA: 3.233	BAL: 251.5

2021 Spring Quarter

ORGAN TRANSPORT SYS	BME	121	4.0	C	8.0
THEORY OF MACHINES	ENGRMAE	145	4.0	B-	10.8
COMP MAT STRUCTURES	ENGRMAE	155	4.0	B+	13.2
ELECTRNC OPTIC PROP	MSE	69	4.0	C+	9.2
SR DESIGN PROJ III	MSE	189C	3.0	A-	11.1
Term Totals	ATTM: 19.0	PSSD: 19.0	GPTS: 52.3	GPA: 2.753	BAL: 14.3
Cumulative Totals	ATTM: 223.0	PSSD: 223.0	GPTS: 711.8	GPA: 3.192	BAL: 265.8

INCOMPLETE GRADES:	0	UNITS:	0.0
NR GRADES:	0	UNITS:	0.0
P/NP GRADES:	0	UNITS:	0.0
S/U GRADES:	0	UNITS:	0.0
W GRADES:	0	UNITS:	0.0

GRADE UNITS ATTEMPTED	223.0	GRADE POINTS	711.8	UC GPA	3.192	BALANCE	265.8
TOTAL UNITS PASSED	223.0	UNITS COMPLETED	235.0				

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